A BEGINNER'S GUIDE TO RISK BACKTESTING

VaR Backtesting, Made Understandable

What a breach means, what the three dashboard tests ask, and how to read the results without needing a statistics background.

The whole idea in one sentence: A VaR backtest asks whether losses crossed the risk threshold about as often as expected, and whether those crossings arrived randomly or in clusters.

The three questions

1 Too many hits?

Kupiec checks the count.

2 Hits in clusters?

Christoffersen checks timing.

3 Both together?

Conditional coverage combines them.

What you will learn

Dashboard item	Plain-language question
Kupiec proportion of failures (POF)	Did the model have too many or too few VaR breaches overall?
Christoffersen independence	Did breaches tend to happen close together?
Christoffersen conditional coverage	Does the model get both the breach rate and timing right together?

We will use the values from your dashboard screenshot as a worked example: 99% VaR confidence, a 500-observation rolling window, 2,714 test observations, and 28 breaches. Your positions are hypothetical, and the example is for learning.

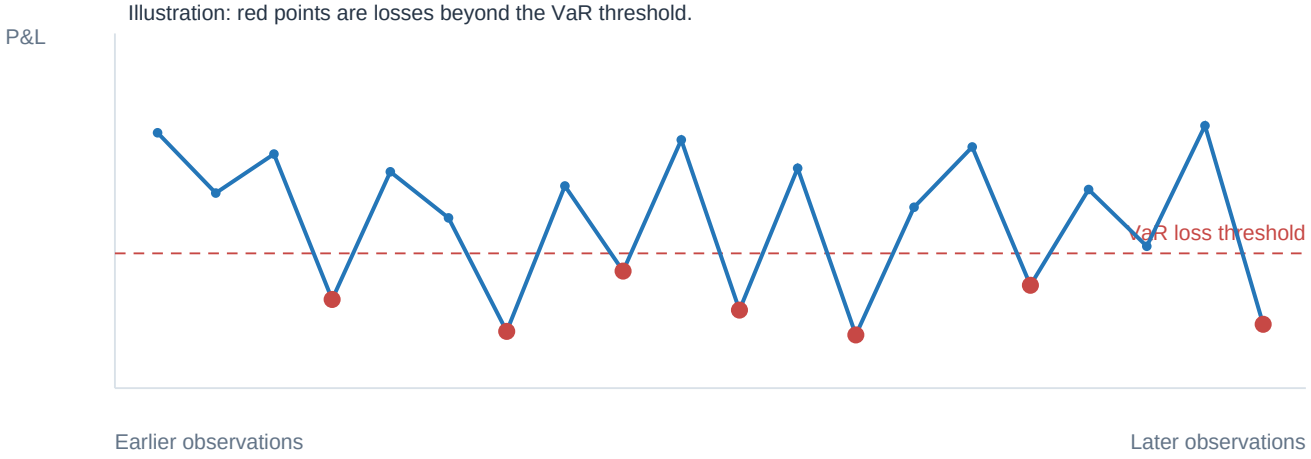
1. Start with VaR and the word 'breach'

Value at Risk (VaR) is a loss threshold for a chosen horizon and confidence. A 99% one-day VaR of $20,000 says: based on the model and data, daily loss is expected to be no worse than $20,000 on about 99% of days. The remaining 1% are tail days when the loss can exceed $20,000.

> VaR is a threshold, not a worst-case limit. A 99% VaR of $20,000 does **not** say that the largest possible loss is $20,000. It says losses larger than that threshold are expected to be rare under the model.

How the plot relates to a breach

Illustration: red points are losses beyond the VaR threshold.



A **breach** (also called an exception or failure in the statistical literature) occurs when the actual loss is larger than the VaR threshold. The word 'failure' does not mean the app crashed; it means the risk forecast was exceeded on that day.

Confidence	Expected breach probability	Expected breaches in 1,000 test days
95%	5%	about 50
99%	1%	about 10
99.9%	0.1%	about 1

The lower the tail probability, the rarer the event. That is why a 99.9% model needs a lot of test observations before we can learn much from its breach count.

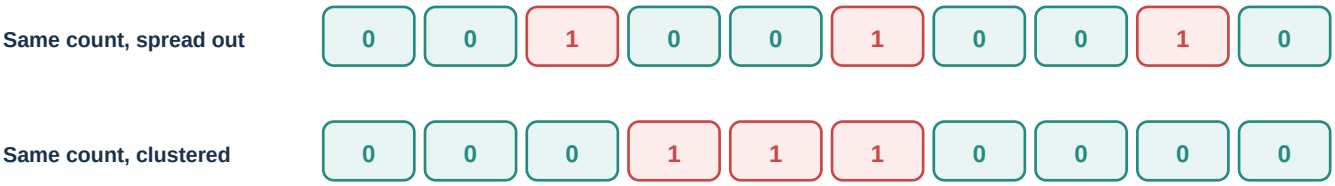
2. What the rolling backtest does

The dashboard does not compare one VaR number against all history. It creates a fresh threshold for each test date using only the observations before that date. For a 500-observation lookback, the forecast for today uses the previous 500 daily portfolio P&Ls. Today's P&L is then checked against that forecast. This keeps the test from using tomorrow's information to predict today.

Step	What happens
1. Look back	Take the previous 500 portfolio P&L observations.
2. Set VaR	Find the loss threshold at the selected confidence, such as 99%.
3. Test the day	Record 1 if the actual loss is beyond that threshold; otherwise record 0.
4. Move forward	Advance one observation and repeat with the latest 500 prior observations.

The backtest becomes a row of 0s and 1s

For each test observation, write **1** for a breach and **0** for no breach. The total number of 1s is the breach count. Their order is the breach pattern.



Both rows above have three breaches. A count-only test sees the same total in each row. A timing test sees that the second row has a run of three adjacent breaches. This difference is exactly why the dashboard shows more than one test.

> In this project, a VaR breach is recorded when **actual P&L is below the negative VaR loss threshold**. A loss of $30,000 breaches a $20,000 VaR; a loss of $15,000 does not.

3. Kupiec POF: did the number of breaches make sense?

POF means **proportion of failures**. In VaR backtesting, a 'failure' is a breach. Kupiec's test ignores the order of breaches and asks one question: **Was the overall breach rate compatible with the target rate?** [1]

At 99% VaR confidence, the expected breach probability is 1%. If you test 2,714 days, the simple expected count is **2,714 x 1% = 27.14 breaches**.

Your screenshot: worked example

Quantity	Value	Meaning
Test observations (n)	2,714	Days with a rolling VaR forecast and an actual P&L.
Observed breaches (x)	28	Actual losses beyond the rolling VaR threshold.
Observed rate	28 / 2,714 = 1.03%	Very close to the 1.00% target.
Kupiec p-value	0.8689	The breach count is not unusually far from the target.

What is the test comparing?

The null hypothesis says the true breach probability is the target (1%). The alternative lets the data choose its own rate (here, 28 / 2,714). Kupiec compares how well those two explanations fit the observed count. A large mismatch gives a large likelihood-ratio statistic and a small p-value.

Interpretation here: p = 0.8689 is much greater than 0.05, so this test does not reject the 1% breach-rate assumption. It does not prove the model is correct, and it says nothing about whether breaches were clustered.

Optional math: the likelihood-ratio statistic compares the likelihood under the target probability p0 with the likelihood under the observed rate p_hat = x / n. Under the null, the statistic is approximately chi-square with 1 degree of freedom. At a 5% significance level, the 1-degree-of-freedom cutoff is about 3.84.

$$LR_{uc} = 2 [x \ln(\hat{p} / p_0) + (n - x) \ln((1 - \hat{p}) / (1 - p_0))]$$

Here p_0 is the target breach probability and $\hat{p} = x / n$ is the observed rate.

4. Christoffersen independence: did breaches cluster?

A VaR model can get the total number of breaches right and still be concerning if breaches bunch together. Several exceptions in a short stressful period can mean the model reacts too slowly to changing risk. Christoffersen's independence test looks at whether today's breach depends on whether the previous observation was a breach. [2]

Previous observation	Next is 0: no breach	Next is 1: breach	How to read
0: no breach	n00	n01	n01 is a new breach after a calm observation.
1: breach	n10	n11	n11 is a breach immediately followed by another breach.

The four transition counts in your screenshot

Transition	Count	Plain meaning
n00 = 2,659	2,659	No breach followed by no breach.
n01 = 26	26	No breach followed by a breach.
n10 = 26	26	Breach followed by no breach.
n11 = 2	2	A breach was immediately followed by another breach twice.

The counts add to 2,713 transitions because 2,714 observations create one fewer adjacent pair. n11 = 2 does not necessarily mean two separate episodes: it could be one run of three breaches (1-1-1), or two separate pairs (1-1 and 1-1). The dashboard's new consecutive-run table helps distinguish those cases.

Optional math: q0 = n01 / (n00 + n01) is the breach chance after a non-breach. q1 = n11 / (n10 + n11) is the breach chance after a breach. Independence means q0 and q1 are the same. The test compares one shared chance with two separate chances; a much better fit from two chances suggests dependence.

Your independence result is LR = 4.533, p = 0.0332. Since p is below 0.05, the test rejects its null hypothesis of independent breach timing at the 5% level. This is evidence of clustering in this backtest sequence, not proof of the cause.

5. Christoffersen conditional coverage: rate and timing together

The conditional-coverage test joins the two questions: **Are breaches happening at the right rate, and are they independent over time?** Its statistic is the sum of the Kupiec coverage statistic and the independence statistic. Its reference distribution has 2 degrees of freedom. [2]

$LR_{CC} = LR_{POF} + LR_{ind}$
For your screenshot: 0.0272 + 4.533 is about 4.5603.

Your screenshot: all three tests side by side

Test	What it asks	LR	p-value	At 5%?
Kupiec POF	Is the breach rate right?	0.0272	0.8689	Do not reject
Independence	Are breach times independent?	4.533	0.0332	Reject
Conditional coverage	Are rate and timing both right?	4.5603	0.1023	Do not reject

Why can independence reject while conditional coverage does not?

They use different reference distributions. The independence test has 1 degree of freedom, so its 5% cutoff is about 3.84; 4.533 is above that cutoff. Conditional coverage has 2 degrees of freedom, so its 5% cutoff is about 5.99; 4.5603 is below that cutoff. The results are not contradictory: the standalone timing test detects a pattern, while the joint test does not find enough combined evidence to reject both conditions at once.

The dashboard's yellow banner is triggered because the independence test rejects. The Kupiec test does not reject, and the conditional-coverage test does not reject. So the specific signal is **possible breach clustering**, not an obviously wrong overall breach frequency.

6. Reading your dashboard result, row by row

The screenshot shows six active hypothetical positions: Brent $500,000, WTI $300,000, Natural Gas $200,000, Coffee $100,000, Bitcoin $100,000, and Ethereum $50,000. The backtest controls are set to 99% confidence and 500 observations. This is a different portfolio from the earlier test with only Natural Gas and ETH, so its P&L and diagnostic results can differ.

Dashboard result	Your value	How to say it in ordinary language
Backtest observations	2,714	We tested 2,714 days after the rolling window was available.
VaR breaches	28	The loss threshold was crossed 28 times.
Observed breach rate	1.03%	28 / 2,714; close to the 1% target.
Expected breach rate	1.00%	At 99% confidence, the model expects a 1% tail rate.
Kupiec p-value	0.8689	The count is not unusual under the 1% target.
Independence p-value	0.0332	The timing test flags clustering at the 5% level.
Conditional coverage p-value	0.1023	The joint test does not reject at the 5% level.

1 Too many hits?

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What conclusion is fair?

The observed frequency is close to its target, and the Kupiec test agrees that the count is not statistically unusual. The independence test flags evidence that breaches are not evenly/randomly spaced. The conditional-coverage test does not reject the joint null at 5%. A careful summary is: **frequency looks reasonable in this sample, but the timing diagnostic raises a clustering question that merits review.**

A low p-value is not the probability that the model is wrong. A high p-value is not the probability that the model is right. See the next section for the exact meaning.

7. P-values, significance, and the two confidence controls

What a p-value means

A p-value asks: **If the test's null assumption were true, how often would random variation produce a result at least this surprising?** A p-value of 0.033 means about 3.3% under that null-and-test setup. It is not a 3.3% chance that the null is true, and it is not a 96.7% chance that the model is bad.

p-value at 5% significance	Decision	Do not say
Below 0.05	Reject this test's null; evidence against that assumption.	The model is definitely broken.
0.05 or above	Do not reject; the test did not find strong enough evidence.	The model passed or is proven safe.

Two percentages in the sidebar, two different jobs

Control	What it changes	Where it is used
Expected Shortfall confidence	Which worst-loss tail is averaged to estimate ES severity.	Portfolio overview ES card.
VaR backtest confidence	The rolling VaR threshold and the expected breach rate (100% - confidence).	Backtest chart, breach metrics, and three tests.

Example: 99% backtest confidence means the VaR model is expected to have about 1 breach in 100 test observations. 99% ES confidence means average the losses at or beyond the 99th-percentile loss cutoff. These controls can be set separately.

Why high confidence needs more data

The dashboard's tail-support note is a rough sample-size guide. A 500-observation window at 99% has about 500 x 1% = 5 nominal tail observations. At 99.9%, a 250-observation window has about 250 x 0.1% = 0.25 nominal observations. That is not a literal quarter of an event: the empirical threshold must use an actual historical order statistic, so it can jump to the single worst observation. Very high confidence with a short lookback makes VaR coarse and unstable.

8. What the tests can and cannot tell you

They can help answer

- Does the backtest have far more or fewer breaches than the selected target? - Do breaches appear to arrive in runs rather than independently? - Which test is producing a warning, and what part of the forecast it questions?

They cannot certify

- That the next loss will stay under VaR or ES. - That all market regimes, products, and future conditions are represented in history. - That a non-rejected test proves the model is correct. - That an illustrative constant-notional portfolio equals a real account with rebalancing, fees, margin, liquidity, or financing.

These likelihood-ratio p-values use large-sample approximations. At very high confidence, expected breach counts can be tiny, making the tests low-powered and their conclusions uncertain. Use the breach count, dates, tail sample size, and market context together.

A simple review routine

Check	Ask yourself
1. Settings	What confidence and lookback are selected? How many nominal tail observations per window?
2. Frequency	Are observed and expected breach rates close? What does Kupiec say?
3. Timing	Do breach dates bunch together? What does independence say?
4. Joint result	Does conditional coverage reject? Read it as its own 2-degree-of-freedom test.
5. Context	Did breaches cluster in a known market shock? Is the tested portfolio the one you intended?

For your current screenshot: 99% / 500 has about 5 nominal tail observations per rolling window. The 1.03% breach rate is close to target. Kupiec does not reject; independence does reject at 5%; conditional coverage does not reject. The next useful action is to inspect the consecutive-breach run dates and losses, then decide whether a regime-aware risk model is needed.

9. Quick glossary and sources

Term	Simple meaning
VaR	A loss threshold associated with a chosen confidence and time horizon.
Expected Shortfall (ES)	Average loss in the selected tail beyond the VaR cutoff.
Breach / exception / failure	An actual loss larger than the VaR threshold.
Backtest observation	A day for which the rolling forecast and actual P&L can be compared.
Lookback	Number of earlier observations used to estimate each rolling VaR threshold.
POF	Proportion of failures: Kupiec's test of the overall breach rate.
Transition	An adjacent pair of outcomes, such as no breach followed by breach.
Null hypothesis	The assumption a test evaluates; rejecting it is evidence against that assumption.
Significance level	A chosen evidence threshold. This dashboard uses 5% (0.05).
Degrees of freedom	A property of the test statistic that determines its reference distribution and p-value.

Sources and project implementation

[1] Paul H. Kupiec (1995), *Techniques for Verifying the Accuracy of Risk Measurement Models*, Federal Reserve Board, Finance and Economics Discussion Series 95-24. FRASER record and paper.

[2] Peter F. Christoffersen (1998), *Evaluating Interval Forecasts*, International Economic Review, 39(4), 841-862. Paper PDF.

Project behavior summarized from **src/risk/backtesting.py**, **src/risk/backtest_summary.py**, and **app/pages/dashboard.py**. Dashboard example values are from the screenshot shared for this discussion. Prepared 23 October 2026.

If you remember only three things: **Kupiec = count**; **Christoffersen independence = clustering**; **conditional coverage = count plus clustering together**.